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Effects of Trial-Adjusted Neurofeedback Training on Motor-Imagery Based Brain-Computer Interface Performance

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***Abstract*—Motor imagery (MI) classification based on electroencephalography (EEG) has been extensively studied and recently used more in brain-computer interfaces (BCI). This study uses left and right hand MI tasks for the BCI system. A common obstacle for MI-BCI is the inability of some participants to perform the BCI task, called BCI illiteracy. Various training protocols have been investigated to improve the performance of BCI but are designed with a balanced dataset. Similarly to how people show a bias towards a side (e.g. left or right) for motor execution tasks, it has been seen that participants also show a performance bias in MI tasks as well. To address this MI bias in participants, a novel neurofeedback protocol was designed to adjust the number of trials each condition has. Trials will be adjusted to increase the number of times participants have to perform their weak MI task. This study aims to investigate the overall effect that the trial-adjusted neurofeedback had on participant's cognitive performance on the MI-BCI system. The effects were investigated through time-frequency and band power analysis. The time-frequency analysis showed improvement in key MI feature and band power analysis results had an improvement on the alpha and beta frequency bands. In the analysis results, trial-adjusted neurofeedback was seen to have an effect on participant's cognitive performance on the MI-BCI task.**

Keywords*—*brain-computer interface, motor imagery, neurofeedback, EEG

I. INTRODUCTION

Brain-computer interfaces (BCI) have led to exciting developments within various applications, making BCI a significant research area in applied neuroscience and neuro-engineering. Non-invasive BCI aids in new methods of neurorehabilitation for people with physical disabilities (e.g., paralysis and amputees patients) and with brain injuries (e.g., stroke patients) [1]. A common obstacle within BCI research is participants that are unable to perform the BCI task, commonly called "BCI illiteracy" [2]. BCI illiteracy has been estimated to affect 15% to 30% of the subject population [3]. Various studies have investigated BCI illiteracy, but the cause is still unknown. While the cause is still being investigated, various training protocols are being designed to help BCI illiterate people become literate in the task. One of the most commonly used training protocols in BCI is neurofeedback (NF). Many of these training protocols are used with electroencephalogram (EEG) based BCI data. As EEG based BCI data is very time expensive and currently has very few openly available datasets, maximizing the amount of usable data is crucial. This is why creating effective training protocols to minimize data loss due to BCI illiteracy is vital.

In this study, trial-adjusted NF is utilized to investigate participant's improvement in the BCI task. Left and right-hand motor imagery (MI) is used as the BCI task. MI is the imagination of a motor action without physically performing it and is commonly used with BCI systems [4]. MI is differentiated between kinesthetic MI (KMI) and visual MI (VMI). KMI is the imagination of the feeling or sensation of the motor action without physically performing it, and VMI is the imagination of the path or movement of the motor action without physically performing it. VMI has a closer similarity to visual observation brain activity, while KMI is similar to motor execution (ME) [4]. KMI was used for the MI task due to its similarity to ME activity. The key difference between ME and KMI are the event-related synchronization/desynchronization (ERS/ERD) within the EEG alpha (8-13 Hz) and beta (13-30Hz) frequency bands. ERS/ERD is an event-related, frequency band specific power increase or decrease. ME have been seen to have an ERD followed by an ERS within the alpha and beta frequency range. In contrast, KMI shows strong correlation with ERD in the alpha and beta frequency range [5].

With MI-BCI, NF was used as the training protocol. While various NF paradigms have been investigated, they all utilized a balanced trial conditions [6]–[9]. We have seen with previous work that participants will express some task performance bias in both ME and KMI [10], [11]. As the reason why people have a MI performance bias is currently not well known, we proposed a NF paradigm that takes this task performance bias effect into consideration. The number of trials for each conditions during NF are adjusted to address the possibility of biased task performance. Similar to the difference in ease of ME for dominant vs. non-dominant hand, we believe that participants in MI-BCI will have a performance bias between their different task performance. It has also been seen that there is a difference in MI brain activity between left- and right-handed people [10]. Crotti et al. showed that the similarity between left- and right-handed people for MI was bilateral activation in the dominant hand and more lateralized activation in the non-dominant hand [12]. This experiment will investigate how adjusting the number of trials in each conditions to the participants performance bias during NF will affect overall MI-BCI performance.

II. Methods and Materials

A. Participants

The experiment was conducted with 10 subjects (7 males and 3 females) aged from 21 to 25 (M=23.6; SD=0.97). Subjects were divided into their performance bias (4 left strong and 6 right strong) seen in Fig. 1. All participants were reported to be right-handed and were volunteers who gave their written consent to part take in the study. Participants also declared they were not taking medication or other psychoactive substance on a permanent basis and have no history of neurological disorder. All subjects gave written, informed consent to participate in the study and that they were able to voluntarily withdraw from the experiment anytime. This study was approved by the Tokyo Institute of Technology Ethical Review Board (IRB- A22435).

B. Experimental procedure

Before the start of the experiment, the experimenter explained the details of the task and what the participants should be doing during the recording. Participants are then asked to sit on a chair and place their head on a chin rest that will be adjusted to a comfortable height. The participant's head is placed in front of a monitor 50 cm away, at eye level. The participants are also asked to place both hands on the table in front of them. Before the start of the experiment, participants were given a practice session consisting of 5 trials per condition to get familiar with the flow of the experimental cues. The experiment consists of three blocks, train, Neurofeedback, and test (Fig. 2). The train and test block are performed in Processing. The neurofeedback block was performed in Psychopy [13]. The first block is the training block which contains no feedback. The conditions are right and left-hand grasp MI. The MI task is the imagining of the feeling and sensation of opening and closing the hand for the instructed side. MI is cued by an arrow pointing left or right. The cue for the condition lasts one second, followed by MI that varies randomly between the time of 3.5 to 4.5 seconds and ending with an 1.5 second rest. There is a total of 100 trials, with 50 trials per condition.

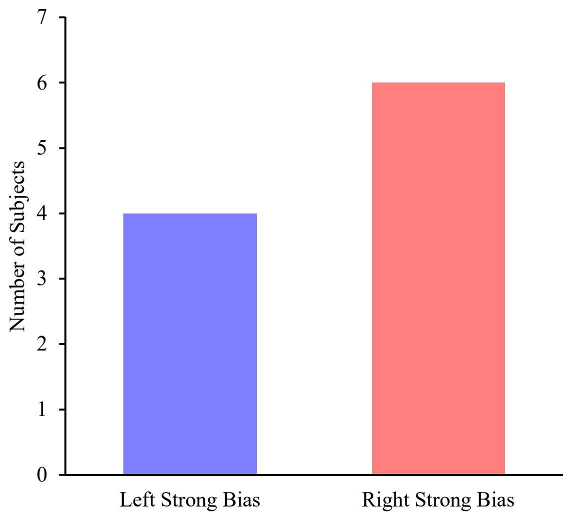

Figure 1. Subjects in each bias group

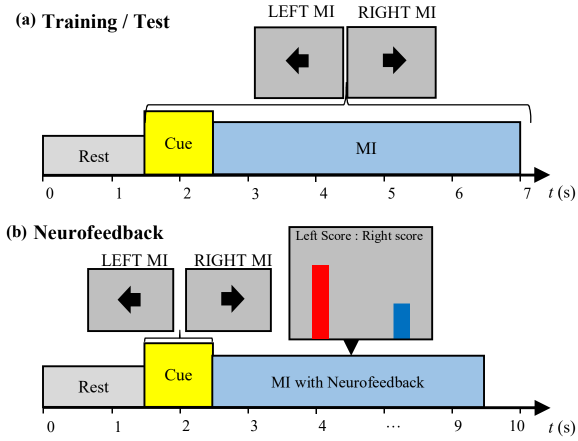

Figure 2. The experiment paradigm is divided into 2 paradigm types. In the training/test block (a), train will be conducted before the neurofeedback and test will be conducted after neurofeedback. In the neurofeedback block (b), the cue will be the same the train/test block. Instead of the fixation cross, neurofeedback is displayed. Two vertically moving bars are displayed on the screen corresponding to the classifier's prediction probability of left hand and right hand MI.

After the first block, offline analysis of the recorded EEG is conducted. Manual inspection of the analysis are used to determine the condition weights for the NF block. The manual inspection involves a simple time-frequency analysis of the channels on the left and right hemisphere of the brain. The key ERD/ERS MI features on alpha and beta were mainly inspected to determine the condition weights. The condition with poor signal quality is weighted higher for more trials during NF. The weights will be distributed as seen in table 1 where left strong indicates that the participant had good feature quality for left hand MI, but weak features for right hand MI and the opposite result for the right strong label. The equal label indicates that the participant had equal feature quality for both left and right hand MI. After the analysis and training of the classifier, the neurofeedback block with the adjusted trials is started. The NF block has the same task as

TABLE I. Subject Neurofeedback Labels

Condition	Left MI Weight	Right MI Weight
Left Strong	0.35	0.65
Equal	0.5	0.5
Right Strong	0.65	0.35

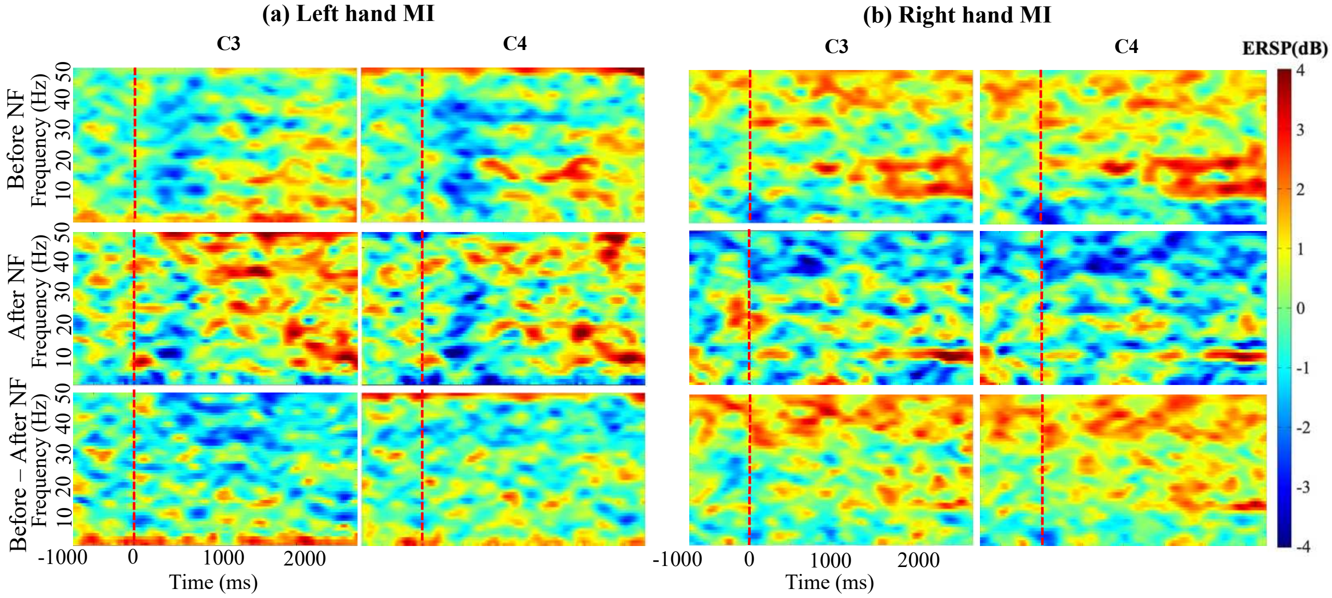

Figure 3. An event-related spectral perturbation (ERSP) plot of subject 9 comparing the blocks before and after neurofeedback within the left and right hand motor imagery task for channels C3 and C4. Subject 9 was in the right bias group. (a) The spectrogram depicts results in left hand MI. The top row shows results the average trials of each task before neurofeedback and the middle show results for the block after neurofeedback. The bottom results depicts the results of the difference between the before and after neurofeedback block. The results of the difference between block will show in the red spectrum if ERD is stronger in the after neurofeedback block and in the blue spectrum if ERS was strong in the after neurofeedback block. (b) Depicts the spectrogram results of right hand MI. All plots are shown within the frequency range of 0.5 to 50 Hz.

the training block with the addition of the feedback on the monitor in replacement to the fixation cross. This block consists of 100 trials with 50 trials per condition, and a two minute break at the halfway point to reduce mental fatigue. The trial starts with a blank screen for 1.5 seconds, followed by a 1 second cue. After the cue, MI is performed for a random time variation of 5, 6, or 7 seconds. The small variation in time was used to help restrict participants from falling into a routine and keep their attention to the task. Lab Streaming Layer (LSL) is used to stream the EEG data to the classifier in real-time and send the feedback data back to the experimental monitor. The feedback on the monitor is two bars that independently grow or shrink according to the classifier's prediction probability during each trial. A sliding window with a window size of 1.5 seconds and a step size of 250 ms is used on the EEG data. The feedback data sent is the average of 5 windows to reduce effects of noise that may cause extreme changes in the prediction probability. After the neurofeedback block, a short break is given, followed by the test block.

The test block is the same as the training block consisting of 50 trials with 25 trials per condition. Recorded data from the training and test block will be analyzed to investigate for any effects on performance from the trial adjusted NF. All three blocks are performed on the same day.

C. *EEG data recording and preprocessing*

EEG was recorded from 8 channels according to the 10-20 international system (FC4, C4, CP4, C6, FC3, C3, CP3, C5). A reference electrode was set on the right earlobe, and the ground electrode was set on the left earlobe. The EEG amplifier used was Cyton Biosensing Board (OpenBCI Inc.) and the signal was recorded with a sampling rate of 250 Hz.

Preprocessing of the EEG data was performed in Python. Common average referencing (CAR) and a Butterworth bandpass filter (0.5-40 Hz) was performed on the data. Sliding window was also performed on each trial with a window size of 1.5 s and step-size of 250 ms. Common spatial pattern (CSP) was used for feature extraction and with a linear discriminant analysis (LDA) classifier [14], [15].

The offline analysis was performed in MATLAB with the EEGLAB toolbox [16]. Signal quality and strength for MI in all conditions were manually inspected to define condition weights.

III. Results

Fundamental analysis was performed for the collected data. There was an apparent difference in EEG signal after one session of the proposed neurofeedback. This was observed through EEG spectral and the band power analysis. ERD/ERS was analyzed in the spectral analysis and a Wilcoxon signed-ranked test was performed on the band power. An α level of 0.05 was considered for statistical significance.

Comparing the results from before to after NF in Fig. 3, an observable difference in the ERSP was seen. Both blocks had an ERD in the alpha and beta frequency range. Comparing the difference between the two blocks, a stronger ERD was observed in the alpha and beta region after NF. This was observed in both the left and right hand condition for both C3 and C4 band power in the alpha and beta bands. As seen in Fig. 4, significant effects where observed between before and after the NF block in left and right hand MI on C3 and C4 .

Differences in effect between left and right strong biases were observed in table 2. Comparing the difference between before and after NF, left strong biased participants had an increase in band power for both alpha and beta. Right strong bias participants showed a decrease in alpha and beta power. Negative difference between before and after NF indicates an increase in band power after NF and a positive difference

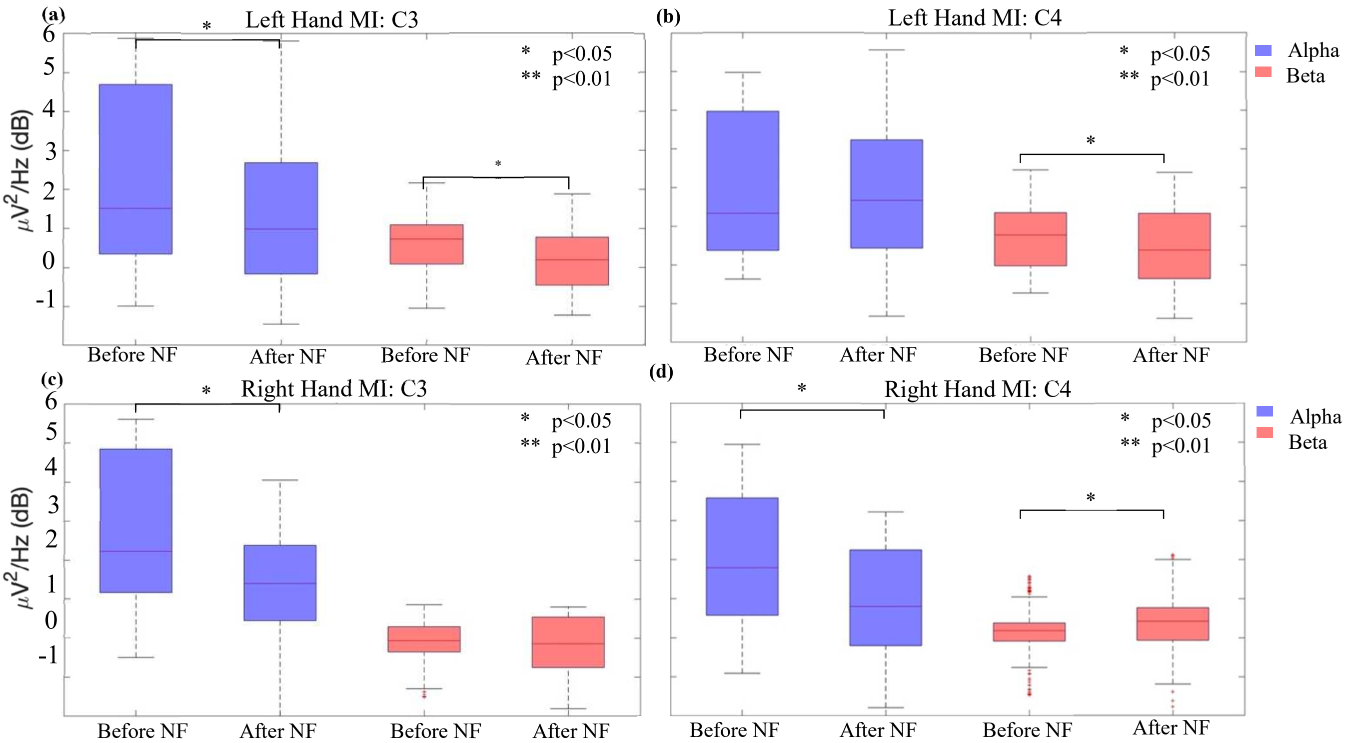

Figure 4. A band power analysis within ERS/ERD of the average of all subject in the alpha- and beta- bands. (a) A boxplot depicting the band power of alpha (blue) and beta (red) for the left hand motor imagery task in channel C3. (b) A boxplot depicting the band power of alpha (blue) and beta (red) for the left hand motor imagery task in channel C4. (c) A boxplot depicting the band power of alpha (blue) and beta (red) for the right hand motor imagery task in channel C3. (d) A boxplot depicting the band power of alpha (blue) and beta (red) for the right hand motor imagery task in channel C4.

indicated a decrease in power after NF. A significant difference in power was seen across both MI condition in the alpha and beta band. Significant difference was seen in left strong bias alpha band: left and right hand MI on C3 and C4. Effects observed between the two bias groups indicates that there was some cognitive effect on participants.

Classification accuracies were also obtained to investigate BCI performance. The average accuracy of the classifier for the before NF block was 69%. After the NF block the average classification accuracy rose to 74%. The mean increase in accuracy was 5%. Increase in accuracy after NF was also seen across all subjects after NF. This showed a positive effect from the proposed NF paradigm.

IV. Discussion

This work proposed a trial adjusted NF paradigm to address MI performance bias between hands in participants. In comparison to a traditional balanced condition NF, weighting the condition trials during the NF block allows participants to have more training time on their poor performed MI task. A full experimental session of train (before NF) and test (after NF) was analyzed with spectral and band power analysis.

Looking into spectral analysis, an observable difference between before and after NF is seen in Fig. 3, where the best performing participant (subject 11) is shown. In Fig. 3a, an improvement MI feature is seen as more ERD is seen for left hand MI on C4 after NF and less in C3. As C4 is on the right hemisphere on the motor area, stronger ERD features were expected to be seen for left hand MI. The "Before - After NF" shows that ERD was slightly stronger in C4 as the difference in ERD in the after NF will be shown in red. Similar results are seen in Fig. 3b, where the ERD features were seen to be stronger in both C3 and C4 after NF.

With band power, the more important frequency band of alpha and beta were investigated [5]. Seen in Fig.3, before and after NF band powers were analyzed. Within alpha-bands, a decrease in power is observed for the after NF block. High power in alpha has been previously seen to indicate higher movement selection demand in motor imagery [17]. The decrease in power of the alpha-band after the NF block may indicate that participants required less movement selection demand for the MI task.

Difference in band power strength between the two biases were also observed. Both the alpha and beta band power was seen to increase after NF in the left strong bias while a decrease in band power after NF was observed in the right strong bias NF as seen in table 2. Increase in band power after NF is indicated with a negative power difference between before and after NF and a positive difference with a decrease in band power after. As left strong bias had an increase in band power it can be inferred that there may have been an overall

TABLE II. Band Power Difference

Bias	Channel	Alpha		Beta	
		Left hand MI	Right hand MI	Left hand MI	Right hand MI
Left Strong	C3	-1.022	-0.106	-2.007	-0.323
	C4	-0.907	-0.007	-1.778	-0.357
Right Strong	C3	0.856	0.455	1.458	0.132
	C4	0.0857	0.268	1.100	-0.182

a. Power difference is the difference of Before and After NF.

increase in movement selection demand while right strong bias had an overall decrease in demand. As all participants were right-handed, this difference in effect of the band power may be due to the difference in participant's internal motor representation. Within left- and right-handed participants, differences in neural activation for MI on dominant and non-dominant hand tasks have been observed [12], [18]. This supports the observed difference in internal representation of motor tasks. As a significant effect was observed on both the left and right strong bias groups, we can infer that the trial-adjusted NF had an effect on the participants.

Comparing the difference in accuracy between before and after NF, an improvement in BCI performance was seen. All subjects had a positive increase in classification accuracy. Although only a 5% increase in accuracy was seen, this improvement was achieved after one session. Improvement in accuracy across all participants provided evidence of the effectiveness that trial-adjusted NF had on the participants. The proposed trial-adjusted NF showed signs of having a strong positive effect on BCI performance.

V. Conclusion

The proposed trial adjusted neurofeedback appeared to have affected the subject's performance on the MI-BCI system. Spectral and band power analysis on alpha and beta frequency band regions were used to investigate the trial-adjusted NF's effects. With spectral analysis, participants were seen to have an improvement in ERD/ERS features. While ME tasks are expected to have contralateral ERD followed an ERS, MI is expected to have contralateral ERD. Within participants, there were improvements in strength of ERD contralaterally in both the alpha- and beta-bands. This showed signs that participants were able to improve their cognitive strategy on how to perform MI.

With band power, significant difference was observed when comparing the before NF and after NF block. Decrease in the alpha power have been seen have correlation to lower movement selection demand [17]. Difference in band power between before and after NF in both biases indicates that participants may also have different motor representations. As a significant overall effect on the band power was observed for both bias groups, the proposed trial-adjusted NF can be interpreted to have some significant effect on participants.

Classification accuracy also had an improvement after the trial-adjusted NF. A mean increase in accuracy of 5% was observed. All participants also had an increase in classification accuracy after NF. This showed signs of positive effects on the participant's BCI performance.

Although a significant difference was seen with an increase in classification accuracy between the before and after NF block, no conclusive statement can be made with only one session of NF training. While there was a positive effect on the participant's performance, further sessions will be need to be run to determine the full effects on the proposed NF. A long term study over multiple experimental session will be required to identifying the learning rate this paradigm has on participants and its overall effectiveness on the performance. The obtained results can further implement more efficient training protocols to increase the number of participants that can utilize BCI systems while decreasing the overall training time of the system.

References

[1] R. Abiri, S. Borhani, E. W. Sellers, Y. Jiang, and X. Zhao, "A comprehensive review of EEG-based brain–computer interface paradigms," J. Neural Eng., vol. 16, no. 1, p. 011001, Jan. 2019, doi: 10.1088/1741-2552/aaf12e.

[2] C. Vidaurre and B. Blankertz, "Towards a Cure for BCI Illiteracy," Brain Topogr., vol. 23, no. 2, pp. 194–198, Jun. 2010, doi: 10.1007/s10548-009-0121-6.

[3] T. Dickhaus, C. Sannelli, K.-R. Müller, G. Curio, and B. Blankertz, "Predicting BCI performance to study BCI illiteracy," BMC Neurosci., vol. 10, no. Suppl 1, p. P84, Jul. 2009, doi: 10.1186/1471-2202-10-S1-P84.

[4] Y. J. Yang, E. J. Jeon, J. S. Kim, and C. K. Chung, "Characterization of kinesthetic motor imagery compared with visual motor imageries.," Sci. Rep., 002 2021, doi: 10.1038/s41598-021-82241-0.

[5] G. Pfurtscheller and F. H. Lopes da Silva, "Event-related EEG/MEG synchronization and desynchronization: basic principles," Clin. Neurophysiol., vol. 110, no. 11, pp. 1842–1857, Nov. 1999, doi: 10.1016/S1388-2457(99)00141-8.

[6] K. K. Ang et al., "Clinical study of neurorehabilitation in stroke using EEG-based motor imagery brain-computer interface with robotic feedback," Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. IEEE Eng. Med. Biol. Soc. Annu. Int. Conf., vol. 2010, pp. 5549–5552, 2010, doi: 10.1109/IEMBS.2010.5626782.

[7] Á. Barbero and M. Grosse-Wentrup, "Biased feedback in brain-computer interfaces," J. NeuroEngineering Rehabil., vol. 7, no. 1, p. 34, Jul. 2010, doi: 10.1186/1743-0003-7-34.

[8] M. Alimardani, S. Nishio, and H. Ishiguro, "Effect of biased feedback on motor imagery learning in BCI-teleoperation system," Front. Syst. Neurosci., vol. 8, 2014, Accessed: Mar. 18, 2022. [Online]. Available: https://www.frontiersin.org/article/10.3389/fnsys.2014.00052

[9] "Two Brains, One Game: Design and Evaluation of a Multiuser BCI Video Game Based on Motor Imagery | IEEE Journals & Magazine | IEEE Xplore." https://ieeexplore.ieee.org/abstract/document/6400237 (accessed Jan. 13, 2023).

[10] D. Zapała et al., "The effects of handedness on sensorimotor rhythm desynchronization and motor-imagery BCI control," Sci. Rep., vol. 10, no. 1, Art. no. 1, Feb. 2020, doi: 10.1038/s41598-020-59222-w.

[11] D. Zapała et al., "The Impact of Different Visual Feedbacks in User Training on Motor Imagery Control in BCI," Appl. Psychophysiol. Biofeedback, vol. 43, no. 1, pp. 23–35, Mar. 2018, doi: 10.1007/s10484-017-9383-z.

[12] M. Crotti, K. Koschutnig, and S. C. Wriessnegger, "Handedness impacts the neural correlates of kinesthetic motor imagery and execution: A FMRI study," J. Neurosci. Res., vol. 100, no. 3, pp. 798–826, 2022, doi: 10.1002/jnr.25003.

[13] J. Peirce et al., "PsychoPy2: Experiments in behavior made easy," Behav. Res. Methods, vol. 51, no. 1, pp. 195–203, 2019, doi: 10.3758/s13428-018-01193-y.

[14] P. Gaur, H. Gupta, A. Chowdhury, K. McCreadie, R. B. Pachori, and H. Wang, "A Sliding Window Common Spatial Pattern for Enhancing Motor Imagery Classification in EEG-BCI," IEEE Trans. Instrum. Meas., vol. 70, pp. 1–9, 2021, doi: 10.1109/TIM.2021.3051996.

[15] Y. Wang, S. Gao, and X. Gao, "Common Spatial Pattern Method for Channel Selelction in Motor Imagery Based Brain-computer Interface," in 2005 IEEE Engineering in Medicine and Biology 27th Annual Conference, pp. 5392–5395., Jan. 2005, doi: 10.1109/IEMBS.2005.1615701.

[16] A. Delorme and S. Makeig, "EEGLAB: an open source toolbox for analysis of single-trial EEG dynamics including independent component analysis," J. Neurosci. Methods, vol. 134, no. 1, pp. 9–21, Mar. 2004, doi: 10.1016/j.jneumeth.2003.10.009.

[17] L. Brinkman, A. Stolk, H. C. Dijkerman, F. P. de Lange, and I. Toni, "Distinct Roles for Alpha- and Beta-Band Oscillations during Mental Simulation of Goal-Directed Actions," J. Neurosci., vol. 34, no. 44, pp. 14783–14792, Oct. 2014, doi: 10.1523/JNEUROSCI.2039-14.2014.

[18] D. Zapała, P. Iwanowicz, P. Francuz, and P. Augustynowicz, "Handedness effects on motor imagery during kinesthetic and visual-motor conditions," Sci. Rep., vol. 11, no. 1, Art. no. 1, Jun. 2021, doi: 10.1038/s41598-021-92467-7.